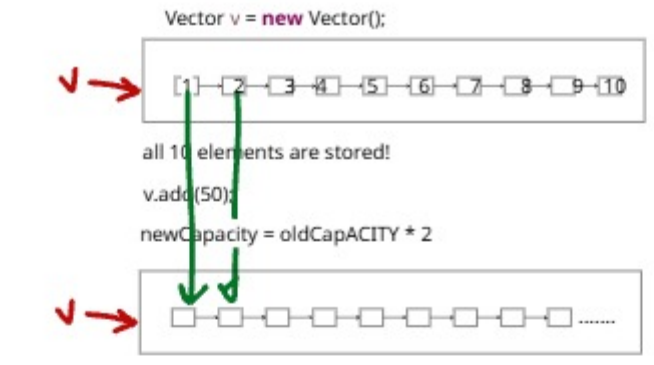


Various methods related to Vector class:-

```
Vector v = new Vector();

    v.add(10);
    v.add(20);
    v.add(30);
    v.add(56.34);
    v.add("Raju");
    v.add(true);
    v.add(10);

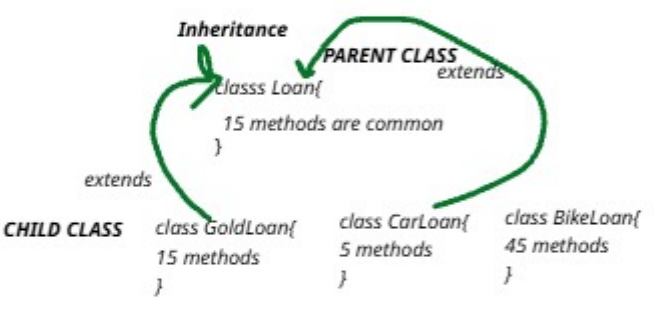
    System.out.println(v.get(1));

    System.out.println(v.contains(30));

    v.set(4, "Ramu");

    System.out.println(v);

    System.out.println(v.capacity());
    System.out.println(v.size());
```



Inheritance is a mechanism where on java class tries to accquire the methods and variables coming from another class

Inherited methods ==> The methods which are coming from parent class directly to the child class, those methods are called as Inherited Methods

Specilaized methods ==> The methods that are present only in the child class which are not inherited from any parent class

```
class A
{
   String name = "Raju";

   public void first() // declaration
   {
      System.out.println("Hello first");// body
   }

   public void second() // declaration
   {
      System.out.println("Hello second");// body
   }
}

class B extends A{

//   2 inherited methods(first and second)
//   specialized method(third)
   public void third()
   {
      System.out.println("Hello third");
   }
}

class C extends A{
//   2 inherited methods(first and second)
}

public class First {

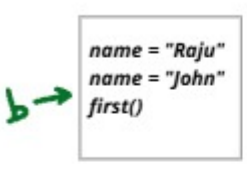
   public static void main(String[] args) {

      B b = new B();
      System.out.println(b.name);
      b.first();
      b.second();

   }
}
```

Only methids can be overriden, but variables cannot be overriden!

```
B b = new B();
   System.out.println(b.name);
   b.first();
```



variable shadowing

It is a mechanism where the child class after inhertieing the avriable friom the parent class, that variable will act like a shadow(backward)

super keyword

It is a keyword in java that will help you to get the parent class content(parent class variable / parent class methods / paremt class constructor)

super keyword should always be used inside the class only

Every java class, comes with a constructior ==> DEFAULT CONSTRUCTOR

Every default constructor will have a super keyword in it!

```
class A
{
   public A()
   {
      System.out.println("A class constructor");
   }

   String name = "Raju";

   public void first()
   {
      System.out.println("A class method!");
   }
}

class B extends A
{

   public B()
   {
      super();
   }

   // specilaized
   public void display()
   {
      System.out.println(super.name);
      super.first();
   }
}

public class First {

   public static void main(String[] args) {

      B b = new B();
      b.display();

   }
}
```

to acces parent class variable ==> super.varname
to acces parent class method ==> super.methodname
to acces parent class constructor ==> super()